

Sine and cosine functions



1

a $\frac{dy}{dx} = -3 \sin 3x$

c $\frac{dy}{dx} = 3 \sin x$

e $\frac{dy}{dx} = -3 \cos x$

g $\frac{dy}{dx} = 20x^4 \cos(4x^5 + 3)$

i $\frac{dy}{dx} = 4x^3 \cos(x^4 + 3)$

k $\frac{dy}{dx} = \frac{1}{3} \cos\left(\frac{x}{3}\right) + 2 \sin x$

m $\frac{dy}{dx} = \cos\left(\frac{t}{4}\right) - 12 \sin 4t + 4t^3$

b $\frac{dy}{dx} = 7 \cos 7x$

d $\frac{dy}{dx} = \cos x - 3 \sin x$

f $\frac{dy}{dx} = 6x^5 \cos(x^6 + 5)$

h $\frac{dy}{dx} = 3 + 5 \cos 5x$

j $\frac{dy}{dx} = -6x \sin(3x^2)$

l $\frac{dy}{dx} = 5x^4 \cos(x^5)$

2

a $\frac{dy}{dx} = 4(\cos x - x \sin x)$

b $\frac{dy}{dx} = \cos 2x$

c $\frac{dy}{dx} = 18 \sin^5(3x) \cos 3x$

d $\frac{dy}{dx} = -\sin 4x \sin x + 8 \cos 4x + 4 \cos x \cos 4x$

e $\frac{dy}{dx} = -\frac{2 \sin 4x}{\sqrt{\cos 4x}}$

f $\frac{dy}{dx} = \sin 2x$

g $\frac{dy}{dx} = 5 \sin 10x$

h $\frac{dy}{dx} = \frac{(x-8) \cos x - \sin x}{(x-8)^2}$

i $\frac{dy}{dx} = -5x^4 \sin 5x - 15 \sin 5x + 4x^3 \cos 5x$

j $\frac{dy}{dx} = 6 \cos 4x \cos 6x - 4 \sin 6x \sin 4x$

k $\frac{dy}{dx} = -2x^{\frac{7}{4}} \sin 2x + \frac{7}{4}x^{\frac{3}{4}} \cos 2x$

l $\frac{dy}{dx} = -12 \sin 4x + 30 \cos 5x$

m $\frac{dy}{dx} = \frac{3(\sin x - x \cos x)}{\sin^2(x)}$

n $\frac{dy}{dx} = 4x^3 \sin\left(\frac{x}{2}\right) + \frac{x^4 \cos\left(\frac{x}{2}\right)}{2}$

o $\frac{dy}{dx} = \sin x + \cos x$

p $\frac{dy}{dx} = 20 \cos 10x$



Applications

3 $y' = \frac{5x \cos 5x - \sin 5x}{x^2}$

4 $\frac{dy}{dx} = \frac{1}{\cos^2(x)}$

5 $\sin^2 x + \cos^2 x = 1$ for all x . Therefore function y is a constant function for all x and has a gradient of 0.

6 a D b B c A, C, E

7 a $\frac{d}{dx}(\sin x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$
 b $\frac{d}{dx}(\cos x) = -x + \frac{x^3}{3!} - \frac{x^5}{5!} + \frac{x^7}{7!} - \frac{x^9}{9!} + \dots$
 c $\frac{d}{dx}(\sin x) = \cos x, \frac{d}{dx}(\cos x) = -\sin x$